
ME 307: Energy Conversion Devices

Course Code	ME 307
Course Name	Energy Conversion Devices
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solids
Intended for	B.Tech.
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- **Thermodynamics, Thermal power plants**
Gas and steam power cycles, Regenerative and reheat cycles
- **Turbo Machinery**
Classification Similitude and specific speeds, Euler turbine equation, Velocity triangles, Turbine and compressor cascades.
- **Axial-flow turbines and compressors**
Stage efficiency and characteristics, Radial equilibrium, Governing
- **Fans, blowers and compressors**
Slip factor, performance characteristics
- **Hydraulic Machines** Pelton wheel, Francis and Kaplan turbines, Draft tubes, Pumps, Cavitation, Fluid coupling and torque converter
- **Introduction to IC engine**
- **Use of Computer Aided Engineering (CAE) in turbomachinery design**

Text Books

1. S.L. Dixon, **Fluid Mechanics, Thermodynamics of Turbomachinery**, 3rd Edition, Pergamon Press, 1998.
2. S M. Yahya, **Turbines Compressors And Fans**, 4th Edition, 2010.
3. B. K. Venkanna, **Fundamentals Of Turbomachinery**, 2009.